

Journal Review

Stratified Thermal Energy Storage (STES): Concept, Design and Significance in Energy Conservation

Abstract

Stratified Thermal Energy Storage (STES) is an efficient method of storing thermal energy by maintaining temperature layers within a storage tank. The technique improves energy utilization in solar thermal systems, district heating, and industrial waste heat recovery. This review discusses the working principle, design aspects, influencing parameters, advantages, and its significance in energy conservation. The study highlights how STES improves system efficiency and reduces auxiliary energy consumption compared to conventional mixed storage systems.

1. Introduction

Thermal energy storage (TES) plays a crucial role in renewable energy systems, especially in solar thermal applications where energy availability is intermittent. Among TES methods, Stratified Thermal Energy Storage (STES) is widely adopted because it preserves temperature layers inside the storage tank without mixing.

Unlike conventional storage tanks where hot and cold water mix, STES maintains distinct thermal layers, improving the quality (exergy) of stored energy.

2. Concept of Stratified Thermal Energy Storage

STES operates based on the principle of density difference. Hot water has lower density and rises to the top, while cold water with higher density remains at the bottom.

Key Concept:

Hot water → Upper layer

Cold water → Lower layer

Thermocline → Transition layer between hot and cold regions

This natural stratification minimizes mixing and maintains higher temperature at the top for efficient utilization.

3. Structure of STES System

A typical STES system consists of:

Cylindrical insulated storage tank

Inlet and outlet diffusers

Temperature sensors

Charging and discharging pipelines

Working Process

Charging Mode:

Hot fluid from solar collectors enters at the top.

Cold return water exits from the bottom.

Discharging Mode:

Hot water is drawn from the top.

Cold water enters from the bottom.

This arrangement preserves temperature layers.

4. Working Principle

The performance of STES depends on:

Buoyancy effect

Low turbulence flow

Proper inlet design (diffusers)

High insulation

Mathematically, stratification efficiency depends on:

Temperature gradient

Reynolds number (flow turbulence)

Heat loss coefficient

5. Types of Stratified Thermal Storage

Water-based sensible heat storage

Molten salt stratified tanks

Underground stratified storage (Aquifer TES)

Applications are seen in:

Solar water heating

District heating systems
Industrial process heat
Combined heat and power (CHP) plants

6. Factors Affecting Stratification

Inlet flow velocity

Tank height-to-diameter ratio

Heat losses

Mixing due to turbulence

Withdrawal rate

Proper diffuser design significantly enhances stratification efficiency.

7. Significance in Energy Conservation

STES contributes to energy conservation in the following ways:

7.1 Improved Solar Utilization

Higher temperature availability at the top improves collector efficiency.

7.2 Reduced Auxiliary Heating

Better stratification reduces backup heater usage.

7.3 Higher Exergy Efficiency

Energy quality is preserved by avoiding mixing.

7.4 Lower Pumping Power

Optimized flow reduces energy consumption.

7.5 Integration with Renewable Systems

STES enhances performance of:

Solar thermal systems

Heat pump systems

Waste heat recovery systems

8. Advantages

Simple design

Low maintenance

Cost-effective

High thermal efficiency

Long service life

9. Limitations

Mixing due to improper design

Heat losses over long duration

Requires large space for high capacity

Performance sensitive to flow rate

10. Case Studies

10.1 District Heating in Hamburg

Large stratified tanks are used to store surplus renewable heat, reducing fossil fuel usage.

10.2 Solar Thermal Plants in Denmark

Denmark widely uses stratified storage in community solar heating plants, achieving high seasonal storage efficiency.

11. Future Research Directions

CFD modeling for thermocline stability

Nano-fluid enhancement

Integration with Phase Change Materials (PCM)

Underground large-scale STES

12. Conclusion

Stratified Thermal Energy Storage is a highly efficient and economical solution for storing thermal energy in renewable systems. By maintaining temperature layers, STES increases energy utilization efficiency and reduces auxiliary power consumption. Its role in solar thermal plants, district heating, and industrial heat recovery makes it a significant technology for sustainable energy conservation.